

ML Strategy Analysis: CNS Oligonucleotide Toxicity

Which theories to prioritize, what the datasets look like, where to extract them, and what to build first.

1. The Analytical Opportunity

CNS oligonucleotide toxicity is a near-ideal substrate for machine learning, for five reasons that almost no other neuropharmacology subfield combines:

1. **Discrete, well-defined input space.** An oligonucleotide candidate is fully describable by ~30 features: a sequence (16-22 nucleotides × 4 bases), a per-position chemistry vector (6-8 modification types), a backbone vector (PS vs PO at each linkage), and a formulation/dose context. Unlike small molecules, there's no "novel scaffold" surprise — the universe of designs is enumerable.
2. **Mechanistic theories with falsifiable predictions.** Unlike most safety pharmacology, where toxicity is "emergent," the field has converged on 4-5 specific molecular mechanisms (cation chelation, paraspeckle binding, RNase H off-target, TLR9, hybridization-dependent off-target). Each predicts which features should matter and which shouldn't.
3. **Quantitative readouts at multiple scales.** From in vitro Ca^{2+} oscillations (continuous) to behavioral scores (ordinal) to NfL biomarker (continuous) to clinical outcome (binary/survival). You can train at one scale and validate at another.
4. **Recent landmark papers with downloadable supplementary data.** Hagedorn 2022 (148 compounds), Moazami 2024 (large chemistry/buffer matrix), Yamaguchi 2025 (late-onset model). Three single papers provide the backbone for a real ML project.
5. **A clear translational endpoint.** NfL is now a regulatory biomarker. Any model that predicts NfL trajectory has direct clinical value, not just academic interest.

The opportunity is **not** to discover new biology — Khvorova, Crooke, and Hagedorn have already done that. The opportunity is to **(a) build the first quantitative predictive model that integrates these mechanisms**, and **(b) generate hypotheses for the parts of design space nobody has explored yet**.

2. Prioritization Framework

Before ranking theories, the criteria:

Criterion	What it means	Weight
Evidence quality	How many independent labs/studies confirm the relationship	High
Effect size	How big is the X→Y change? (binary rescue vs noisy correlation)	High
Dataset availability	Can you actually get the data, or is it locked in pharma?	Critical
Tractability	Can a typical ML model handle the data shape?	Medium
Clinical relevance	Does this connect to a real-world endpoint?	High
Novelty	Has someone already published the ML model?	Medium

A theory wins on this framework if it's well-evidenced, has accessible data, a large effect size, and hasn't already been ML-modeled to saturation.

3. Priority Theories — Ranked

Tier 1: Build the model on these first

Theory 1: Sequence + chemistry → acute CNS tolerability (Hagedorn class)

The claim: Acute tolerability in mouse ICV is largely predictable from primary sequence features (3'-G density, total G count, CpG count, motif content) and chemistry vector (PS positions, sugar at each position).

Evidence: Hagedorn 2022, *Nucleic Acid Ther* — 148 LNA gapmers, ~40% failed acute tolerability, with a published random-forest/logistic predictor that achieves ~80% accuracy. Independent confirmation in Moazami 2024 across chemistry classes.

Why prioritize: The most validated tox relationship in the field; the published Hagedorn dataset is the only large-scale public CNS tox-with-sequences dataset in existence. Anyone serious about CNS-ASO ML must beat or extend Hagedorn's model.

ML framing: Classification (tolerable vs not) or regression on 0–3 behavioral score.

Theory 2: PS content dose-response

The claim: Acute motor phenotype severity scales monotonically with number of PS linkages in the oligo, controlling for sequence and chemistry. Mixed PS/PO backbones are universally better than fully-PS at the same dose.

Evidence: Moazami 2024, *Mol Ther* — quantitative behavioral scoring across DNA/2'-OMe/MOE/LNA sugar analogs with varying PS counts. Replicated in sheep.

Why prioritize: Cleanest dose-response relationship in the literature; high-effect-size; should be the “sanity check” that any model picks up as a top-3 feature.

ML framing: Regression of phenotype score on PS count + sequence + chemistry + buffer interactions.

Theory 3: Hybridization-independent protein binding → late-onset toxicity

The claim: Late-onset (≥ 3 days) neurotoxicity is driven by PS-ASO binding to paraspeckle proteins P54nrb/PSF, nucleolar mislocalization, p53 activation, and apoptosis. A 2'-O-methoxy at gap position 2 (Crooke) or a 5'-cyclopropylene at certain gap positions (Yamaguchi/Obika) abolishes the toxicity without losing knockdown.

Evidence: Crooke lab — multiple papers 2018–2022 with biochemical evidence; Yamaguchi 2025 in vivo rescue in mouse and rat.

Why prioritize: This is the most mechanistically rich, position-specific relationship in the field; ML can extract position-specific rules from sequence-chemistry tuples that humans might miss. It's also the **newest** (5'-CP paper is 2025), so the ML problem is wide open.

ML framing: Multi-task: predict (a) protein-binding affinity at each gap position from chemistry features, (b) late-onset behavioral phenotype, (c) knockdown efficiency. Optimize for binding $\downarrow$ $\times$ knockdown $\uparrow$.

Tier 2: High-value but harder to assemble data

Theory 4: Divalent cation in formulation → seizure incidence

The claim: Adding $\text{Ca}^{2+}/\text{Mg}^{2+}$ to the injectate at molar excess of PO/PS bonds abolishes acute seizures regardless of sequence or chemistry.

Evidence: Khvorova 2024 — binary effect across multiple ASO/siRNA candidates in mice.

Why prioritize: Largest effect size in the entire field — close to “switch on/off.” A model that

captures this needs to encode formulation context, which most ASO ML models don't.

Why Tier 2: The dataset is small (one paper, maybe 20-30 conditions). Useful as a validation rather than a primary training set.

ML framing: A simple interaction term `[Ca2+]formulation × bond_count_oligo` should dominate predictions if the model is properly featurized.

Theory 5: CpG/TLR9 activation → innate immune response

The claim: CpG dinucleotides in PS-containing oligos engage TLR9, producing NF-κB activation and cytokine release. Sugar modification of the CpG (2'-OMe on C or G) abolishes engagement.

Evidence: Yasuhara 2024; older Krieg lab CpG literature; cross-confirmed across multiple cell lines and species.

Why prioritize: Most interpretable mechanism, with clear biochemistry. CpG counting is trivial; the value of ML is in predicting *which* CpG positions matter most given chemistry context.

ML framing: Regression on NF-κB reporter signal or cytokine fold-change; features = CpG positions × chemistry vector at those positions.

Theory 6: NfL trajectory → clinical outcome

The claim: Plasma/CSF NfL change in the first 3-6 months of dosing predicts clinical outcome. Reduction predicts benefit (tofersen, ION363); elevation predicts harm (tominersen, branaplam).

Evidence: Tofersen approval was built on this; ION363 case series; tominersen failure analysis.

Why prioritize: Direct clinical/regulatory relevance — a model here has real-world deployment value.

Why Tier 2: The dataset is patient-trajectory data, mostly aggregated in published trials, not openly available row-by-row. Requires either trial-level meta-analysis or industry partnership.

ML framing: Survival analysis or longitudinal regression; could use Cox or transformer architectures depending on data depth.

Tier 3: Worth doing eventually

- **Hybridization-dependent off-target prediction** — well-studied for liver (Burdick,

Burel); CNS-specific data sparser

- **Stereochemistry-dependent toxicity** (Wave Sp/Rp) — proprietary, hard to assemble
- **Delivery-route / dose-frequency optimization** — clinical-only data

4. Dataset Architecture (per Theory)

For each top-tier theory, here's what the training table should look like.

Dataset A: Sequence + Chemistry → Acute Tolerability (Theories 1 + 2)

Row = one oligonucleotide candidate tested in vivo Suggested size to be useful: ~150–500 rows minimum; ~1000+ ideal

Column	Type	Source	Notes
oligo_id	str	assigned	Unique identifier
sequence	str (16–22 nt)	paper supplement	5'→3' sequence
length	int	derived	
gc_pct	float	derived	Total GC%
g_count	int	derived	Total guanines
g_3prime_5nt	int	derived	G count in 3'-most 5 nt — Hagedorn predictor
cpg_count	int	derived	Number of CpG dinucleotides
gq_score	float	computed	G-quadruplex propensity (G4Hunter or similar)
chemistry_vector	list[str]	paper	Per-position modification: DNA/OMe/MOE/F/LNA/cEt/CP
ps_count	int	derived	Number of PS linkages
ps_positions	list[int]	paper	Indices of PS linkages
gapmer_arch	str	paper	E.g., "5-10-5", "3-14-3"

target_gene	str	paper	E.g., MAPT, HTT, SOD1
dose_nmol	float	paper	ICV dose
formulation_ca_mM	float	paper / methods	Free Ca ²⁺ in injectate
formulation_mg_mM	float	paper / methods	Free Mg ²⁺
buffer	str	paper / methods	HEPES / PBS / aCSF / lactate
species	str	paper	mouse / rat / NHP
route	str	paper	ICV / IT
behavior_score_1h	int 0-5	paper Table	Hagedorn/Moazami scoring
behavior_score_24h	int 0-5	paper Table	
behavior_score_7d	int 0-5	paper Table	Late-onset proxy
seizure_observed	bool	paper	EEG/EMG if reported
survival_7d	bool	paper	
knockdown_pct	float	paper	Target mRNA reduction at terminal timepoint
source_paper_doi	str	bookkeeping	For provenance

Primary extraction sources (in order of yield):

1. **Hagedorn 2022, *Nucleic Acid Ther*** — Supplementary Table S1 has all 148 sequences + scores. Single richest public source.
2. **Moazami 2024, *Mol Ther*** — Supplementary spreadsheet across chemistry/buffer matrix; ~30-50 conditions.
3. **Yamaguchi 2025, *Mol Ther Nucleic Acids*** — sequences + late-onset behavioral data with 5'-CP variants.
4. **Tagami/Yokota 2024** — G-clamp/aminoethoxyphenoxazine variants.
5. **Burel et al. 2015, *Nucleic Acids Res*** — high-affinity LNA gapmer hepatotox data

(extrapolatable to CNS framework).

6. **Vickers/Crooke 2019, 2020** — protein-binding datasets correlated with tox.

Dataset B: Position-Specific Chemistry → Late-Onset Toxicity (Theory 3)

Row = one (oligo, gap position, modification) triple

This is harder — you're trying to predict the *position-specific* effect of a chemistry choice.

Column	Type	Source
parent_oligo_id	str	
gap_position	int (1–10 for typical gapmer)	
wildtype_chemistry	str (DNA usually)	
tested_modification	str (OMe/CP/MOE/F)	
protein_binding_fc	float	Crooke/Vickers binding assays
paraspeckle_score	float	NEAT1/P54nrb imaging
p53_induction_fc	float	qPCR P21/MDM2
cell_viability_pct	float	iPSC neuron cytotox
late_onset_score	int 0-3	Yamaguchi behavioral
knockdown_pct	float	
source_paper_doi	str	

Primary extraction: Yamaguchi 2025 supplementary data; Crooke lab series 2018-2022 (multiple papers in *NAR*, *Mol Ther*); Liang et al. 2019, *NAR*.

Dataset C: NfL Trajectory → Clinical Outcome (Theory 6)

Row = one patient-trial-arm-timepoint

Column	Type	Source
trial_id	str	NCT number
drug	str	Tofersen/tominersen/BIB080/etc.

dose_mg	float	trial publication
frequency	str	Q4W/Q8W/Q16W
patient_id	str (anonymized)	trial publication / supplement
disease	str	ALS/HD/AD/Dravet
timepoint_wk	int	
csf_nfl	float	
plasma_nfl	float	
csf_target_protein	float	E.g., SOD1, mHTT, tau
csf_cell_count	int	
csf_protein_mgdL	float	
ventricular_vol_mL	float	
clinical_score	float	ALSFRS-R, cUHDRS, CDR-SB
outcome_at_endpoint	str	improved/stable/worsened/died

Primary extraction: Patient-level data is mostly **not** public. What you can get:

- Trial-level summary statistics from NEJM/Nature Med publications and supplements
- Some patient-level data via Vivli or YODA data-sharing requests (Biogen and Roche both participate)
- AdComm briefing books for tofersen (FDA Advisory Committee documents have detailed safety tables)
- Aggregated trial-arm data is sufficient for many useful analyses

Dataset D: Innate Immune Activation (Theory 5)

Row = one (sequence, chemistry context, cell type) triple

This dataset is the most tractable to **generate** if you have wet-lab access: HEK-Blue-TLR9 cells + NF-κB reporter is a standard 96-well assay.

Column	Type	Source

sequence	str	
cpg_count	int	derived
cpg_chemistry	list[str]	E.g., "[OMe, MOE]" for two CpGs
flanking_chemistry	list[str]	
tlr9_nfkb_fold	float	reporter assay
cell_type	str	HEK-TLR9 / hPBMC / microglia
concentration_uM	float	
source	str	

Primary extraction: Yasuhara 2024 supplement; older Krieg/Vollmer CpG literature (1990s-2000s); InvivoGen TLR9 ligand product sheets for control sequences.

5. Where to Extract — Concrete Source Map

Tier 1 sources (free, immediate, structured)

Source	What's there	Format
PubMed Central	Open-access papers with supplementary data	XML, PDF, sometimes XLSX
bioRxiv	Preprints; often more raw data than published versions	PDF + supplementary downloads
ClinicalTrials.gov	Trial designs, primary outcomes, AE tables (XML API)	Structured
FDA accessdata.fda.gov	Drug labels, Advisory Committee briefing docs, integrated safety reviews	PDF (parseable)
EMA EPAR documents	Equivalent EU regulatory documents	PDF
DrugBank (academic license)	Approved drugs with structures, targets,	Structured

free)	safety	database
CHEMBL	Bioactivity data including some ASO entries	SQL dumps
OpenTargets	Disease-target mappings	API

Tier 2 sources (require curation effort)

Source	What's there	Effort
Paper supplementary tables	The real gold — Hagedorn, Moazami, Yamaguchi, Crooke supplements	Manual extraction; ~hours per paper
Patent filings (USPTO, Google Patents)	Ionis/Alnylam/Wave/Stoke patent applications list sequences + chemistries + sometimes tox data	Major effort; high yield
Conference abstracts	OTS, ASGCT, AAN, MDS Congress posters; often have data not yet published	Manual scraping
GitHub/Zenodo for paper code	Some authors deposit full datasets (rare in ASO field, more common in CRISPR)	Direct download if available

Tier 3 sources (require partnership or request)

Source	What's there	Access
Vivli (vivli.org)	Patient-level data from Biogen, Roche, others	Submit research proposal
YODA Project (Yale)	J&J / pharma patient-level data	Application
Direct correspondence with authors	Many authors will share extended datasets if you ask politely	Email
n-Lorem Foundation	Personalized ASO data; n-of-1 cases	Collaboration
Industry	Ionis/Alnylam have published "platform tox"	Sponsored

Tier 4 sources (you generate the data)

If your “challenge” includes wet-lab or simulation capacity:

- **In vitro Hagedorn Ca^{2+} -oscillation assay** in primary cortical neurons — ~\$50-100/compound, scalable
 - **TLR9-NF- κ B reporter** in HEK-Blue cells — ~\$10/compound, very scalable
 - **In silico off-target BLAST screening** — free, transcriptome-wide
 - **Molecular dynamics of PS-ASO/protein interactions** — compute-intensive but possible (Crooke approach has been simulated)
-

6. The Recommended Starter Project (MVP)

If I were scoping this as a 3-6 month ML project, here’s the build order:

Phase 1: Reproduce Hagedorn (2-4 weeks)

Goal: Build a binary classifier (tolerable vs not) on the Hagedorn 148-compound dataset. Achieve published ~80% accuracy.

Why: Forces you to set up featurization, train/validation splits, and a baseline. If you can't reproduce the published result, your pipeline is wrong.

Featurization:

- One-hot sequence (length $\times$ 4)
- One-hot chemistry per position (length $\times$ n_modifications)
- PS vector (length-1 binary)
- Derived: total G, 3' G-count, CpG count, GC%, length

Model: Start with random forest and gradient boosting (LightGBM). Move to a small transformer only after you have a baseline.

Phase 2: Extend to multiple chemistry classes (1-2 months)

Goal: Merge Hagedorn (LNA-heavy) with Moazami (multi-chemistry) and Yamaguchi (5'-CP) datasets. Train a single model across chemistry classes.

Key question to answer: Does the same sequence rule (3'-G content) apply across chemistries, or are there chemistry-specific tox rules?

Featurization upgrade: Add interaction features (chemistry × position × G-content).

Phase 3: Position-specific late-onset prediction (1-2 months)

Goal: On the Yamaguchi/Crooke data, predict whether a given (sequence, position, modification) tuple will rescue late-onset tox.

This is where ML gives novel insight — the published rules are about ~3 positions; ML can survey all 22.

Model: Multi-task neural network or attention-based architecture.

Phase 4: NfL meta-analysis (parallel track, 2-3 months)

Goal: Curate trial-arm-level NfL trajectory data from all published CNS-ASO trials. Build a model that, given (chemistry class, dose, frequency, disease, baseline NfL), predicts 6-month NfL change.

Why parallel: The data is harder to assemble (mostly manual), but the model is **directly clinically useful** — could inform dose-frequency choice in new trials.

Phase 5: Active learning loop (if you have wet-lab access)

Goal: Use the trained model to propose ~50 new oligo designs in unexplored design space, then test 10-20 of them experimentally and update the model.

This is where you go from "reproducing literature" to "generating science."

7. Theory-by-Theory ML Approach

Theory	Best model class	Reason
Hagedorn sequence prediction	Gradient boosting (LightGBM, XGBoost)	Tabular features, ~150-500 rows, interpretable
Multi-chemistry tox	Transformer with per-position chemistry embedding	Sequence + structured features, scales with data
Late-onset / position-specific	Multi-task NN with attention over positions	Position-specific rules require positional attention

CpG/TLR9 prediction	Simple LR or random forest	Few features, well-defined biochemistry
Divalent cation rescue	Logistic regression with interaction term	Effect is so binary, complex models are overkill
NfL trajectory	Mixed-effects model or longitudinal transformer	Patient-level repeated measures
Off-target hybridization	Sequence alignment + thermodynamic T _m + RF	Standard bioinformatics + ML hybrid

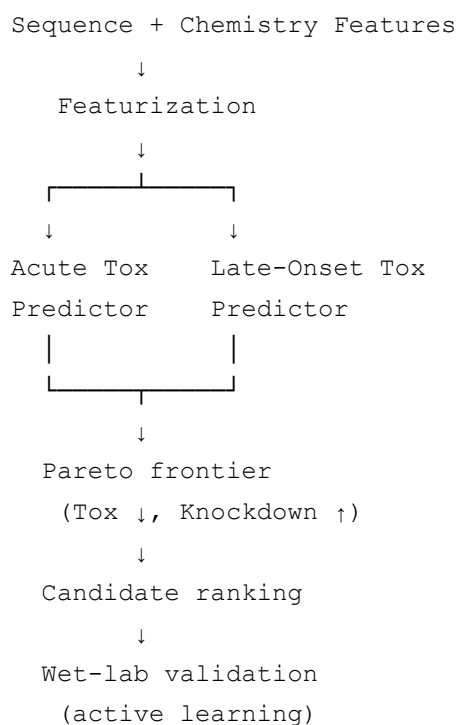
8. Pitfalls & Caveats

1. **Class imbalance** — Most published ASO datasets are enriched for tox-tested candidates (researchers don't publish negative-tox findings). Be careful about base rates; the Hagedorn 40% failure rate is high by design, not natural.
2. **Provenance gaps** — Older papers often don't report formulation details (Ca²⁺ in buffer, exact buffer composition). Treat these as missing-not-at-random; consider sensitivity analyses.
3. **Species translation** — Mouse ICV ≠ human IT. The Khvorova lab shows good mouse→NHP translation; mouse→human is less established. Be cautious about training on mouse-only data and claiming human predictions.
4. **Sequence-chemistry correlation** — In the Hagedorn dataset, certain sequences are tested in only one chemistry context. Be careful about confounded features.
5. **Publication bias** — Patents and conference abstracts contain more negative data than papers. If you skip them, you'll overestimate tolerability.
6. **The "platform" assumption** — Ionis's PS-MOE gapmer dataset is a model class of its own. Don't assume conclusions translate to PMOs, siRNAs, or PN-backbones without validation.
7. **Domain shift between assays** — Behavioral score in mouse ≠ Ca²⁺ oscillation in vitro ≠ NfL in human. Multi-task / multi-readout models help, but treat each readout as a partial signal.
8. **Tominersen-confounded labels** — Late-onset tox in clinical trials confounds chemistry, dose-frequency, on-target effects, and patient biology. Use preclinical data for chemistry inference; reserve clinical data for translational validation only.

9. The Headline Recommendation

If you only do one thing: **build a model that predicts the Hagedorn behavioral score from sequence + chemistry, then extend it with the Moazami and Yamaguchi datasets to cover multi-chemistry and late-onset endpoints.** That model alone — if you achieve AUC > 0.85 on a held-out test set — would be publishable and immediately useful to anyone designing CNS ASOs. It also sets you up for Phases 3-5.

The full pipeline:



10. Provenance & Reusability Notes

For any dataset you build:

- **Use a controlled vocabulary** for modifications (we'd suggest the ONT [Oligonucleotide Therapeutics] ontology or your own version with explicit definitions).
- **Use NCBI standardized notation** for sequences (5'→3', uppercase, no ambiguity).
- **Cite every source paper at the row level** — provenance matters for both reproducibility and for tracking down errors.
- **Deposit on Zenodo or Figshare** with a DOI when published — the field needs more

open datasets, and yours becomes citable.

- **Build a Croissant or HuggingFace dataset card** if you want it widely reused.

End of analysis.